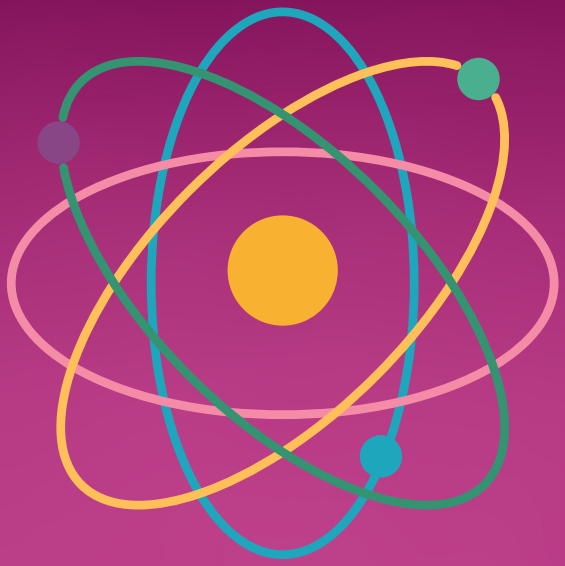
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Chemical Kinetics

English Medium

EXERCISE # S-I

RATE OF REACTION

1. Ammonia and oxygen reacts at higher temperatures as

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$$

In an experiment, the concentration of NO increases by 1.08×10^{-2} mol litre^{-1} in 3 seconds. Calculate.

(i) If rate of reaction is $A \times 10^{-4}$ mol/litre then A will be.

(ii) If rate of disappearance of ammonia $B \times 10^{-4}$ mol/litre then B will be.

(iii) If rate of formation of water $B \times 10^{-4}$ mol/litre then B will be.

CK0214

2. The reaction $2A + B + C \rightarrow D + E$ is found to be first order in A, second order in B and zero order in C. How many times rate of reaction increase if concentrations of A, B, and C becomes double ?

CK0215

3. At 27°C, it was observed during a reaction of hydrogenation that the pressure of hydrogen gas decreases from 20 atmosphere to 11 atmosphere in 75 minutes. Calculate the rate of disappearance of H_2 in M sec^{-1} and in terms of atm min^{-1} are $x \times 10^{-6}$ & $y \times 10^{-3}$ then value of x & y will be

[Use : R = 0.08 atm-L/K-mole]

CK0216

4. For the elementary reaction $2A + B_2 \longrightarrow 2AB$. Calculate how much the rate of reaction will change if the volume of the vessel is suddenly reduced to one third of its original volume?

CK0004

5. For the reaction $3BrO^- \rightarrow BrO_3^- + 2Br^-$ in an alkaline aqueous solution, the value of the second order (in BrO^-) rate constant at 80°C in the rate law for $-\frac{d[BrO^-]}{dt}$ was found to be 0.06 L mol^{-1} s^{-1}. What is the rate constant in L mol^{-1} s^{-1} when the rate law is written for :

(a) $\frac{d[BrO_3^-]}{dt}$, (b) $\frac{d[Br^-]}{dt}$ **CK0217**

6. $xA + yB \rightarrow zC$. If $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = 1.5\frac{d[C]}{dt}$, then (x + y + z) is.

(x, y & z are the lowest integral value.)

CK0006

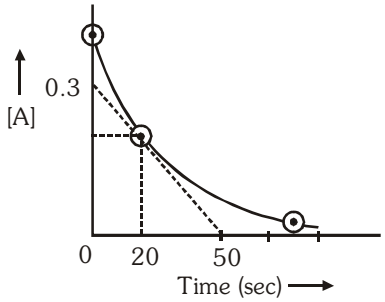
7. For the reaction : 3A (g) → 2B(g), the rate of formation of 'B' at 298 K, is represented as

$ln\left(\frac{d[B]}{dt}\right) = -0.04 + 2 \times ln[A]$. The order of reaction is-

CK0007

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8. A certain reaction **A → B** follows the given concentration (Molarity) – time graph. Calculate the rate in milimolar/sec for this reaction at 20 second.



CK0218

9. The reaction $A(g) + 2B(g) \longrightarrow C(g) + D(g)$ is an elementary process. In an experiment, the initial partial pressure of A & B are $P_A = 0.6$ and $P_B = 0.8$ atm. Calculate the ratio of rate of reaction relative to initial rate when P_C becomes 0.2 atm. **CK0009**

ZERO ORDER REACTIONS

10. In the reaction (A ⟶ B) rate constant is 1.2×10^{-2} M s^{-1} . What is concentration of B (in mole/litre) after 10 and 20 min., if we start with 10 M of A. **CK0010**

11. From the following data for the zero order reaction : A ⟶ products. Calculate the value of k in M min^{-1}.

Time (min.)	[A]
0.0	0.10 M
1.0	0.09 M
2.0	0.08 M

CK0011

12. A drop of solution (volume 0.10 ml) contains 6×10^{-6} mole of H^+. If the rate constant of disappearance of H^+ is 1×10^7 mole $litre^{-1}$ sec^{-1}, If time for complete disappearance for H^+ is $x \times 10^{-10}$ sec, then x will be

CK0219

13. A certain substance A is mixed with an equimolar quantity of substance B. At the end of an hour A is 75% reacted. Calculate the time in hr when A is 10% unreacted. (Given: order of reaction is zero for both A & B.) **CK0013**

14. For a zero order chemical reaction,

$$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$$

rate of reaction = 0.1 atm/sec. Initially only NH_3 is present & its pressure = 3 atm. Calculate total pressure in atm at t = 10 sec. **CK0014**

FIRST ORDER REACTIONS

15. A first order reaction is 75% completed in 72 min. How long time in minute will it take for
(i) 50% completion (ii) 87.5% completion **CK0015**

16. A first order reaction is 20% complete in 10 min. Calculate (i) the specific rate constant min^{-1}, (ii) the time taken min. for the reactions to go to 75% completion. [ln 10 = 2.3, ln2 = 0.7] **CK0016**

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17. Calculate ratio of the time required for 99.9% of the reaction to take place to time required for 50.00% of the reaction to take place if reaction follows first order kinetics.
[Give your answer as nearest integer] **CK0017**

18. A drug is known to be ineffective after it has decomposed 75%. The original concentration of a sample was 500 units/ml. When analyzed 20 months later, the concentration was found to be 250 units/ml. Assuming that decomposition is of I order, what will be the expiry time in month of the drug? **CK0018**

19. A viral preparation was inactivated in a chemical bath. The inactivation process was found to be first order in virus concentration. At the beginning of the experiment, 2.0 % of the virus was found to be inactivated per minute . Evaluate k in min^{-1} for inactivation process. [ln 10 = 2.3, ln 98 = 4.58]

CK0220

20. The reaction $SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$ is a first order gas reaction with $k = \left(\frac{35}{9}\right) \times 10^{-4}$ sec^{-1} at 320°C. What % of SO_2Cl_2 is decomposed on heating this gas for 90 min. (ln2 = 0.7)
CK0020

21. The decomposition of a compound A in solution follow first order kinetics. If 10% w/v solution of A is 10% decomposed in 10 minutes at 10°C, then 20% w/v solution of A is ..... % decomposed in 20 minutes at 10°C. **CK0021**

22. Calculate the half-life in min of the first-order reaction

$$C_2H_4O(g) \longrightarrow CH_4(g) + CO(g)$$

if the initial pressure of $C_2H_4O(g)$ is 80 mm of Hg and the total pressure at the end of 20 minutes is 120 mm of Hg

CK0221

SECOND ORDER REACTIONS

23. In the II order reaction : $2A \rightarrow A_2$. The rate of formation of A_2 is 10^{-5} M sec^{-1} at 0.01M concentration of A. Calculate the rate constant in the rate of disappearance of 'A' in M^{-1} sec^{-1}.

CK0222

24. If $t_{1/2}$ of a second order reaction is 1.0 hr. After what time in hr, the amount of reactant will be 25% of the initial amount?

CK0223

25. Reaction : A + B → C + D, follows the rate law : $r = (2 \times 10^{-3}\ M^{-1}\ s^{-1})\ [A]\ [B]$. The reaction is started with 1.0 mole each of 'A' and 'B', in constant volume of 5 litre. In what time in sec., the moles of 'A' become 0.25 ?

CK0224

DETERMINATION OF ORDER OF REACTION & RATE LAW

26. At 800° C, the rate of reaction, $2\ NO(g) + H_2(g) \longrightarrow N_2(g) + H_2O(g)$
changes with the concentration of NO(g) and $H_2(g)$ as-

Exp.no.	[NO] in M	$[H_2]$ in M	$-\frac{1}{2}\frac{d[NO]}{dt}$ in M sec^{-1}
(i)	1.5×10^{-3}	4×10^{-3}	4.50×10^{-9}
(ii)	1.5×10^{-3}	2×10^{-3}	2.25×10^{-9}
(iii)	3.0×10^{-3}	2×10^{-3}	9.00×10^{-9}

(a) What is the order of reaction?
(b) What is the rate of reaction in M sec^{-1} when $[H_2] = 1.5$ M and [NO] = 1.0M ?

CK0225

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27. The catalytic decomposition of $N_2O(g)$ by gold at 900°C and at an initial pressure of 200 mm is 50% complete in 53 minutes and 73% complete in 100 minutes.

(i) What is the order of the reaction?

(ii) Calculate the velocity constant in $M^{1-n}\,min^{-1}$ here is n is order of reaction.

(iii) How much of percentage $N_2O(g)$ will decompose in 100 min. at the same temperature but at initial pressure of 600 mm of Hg ?

CK0226

28. The pressure of a gas decomposing at the surface of a solid catalyst has been measured at different time and the results are given below

t (sec)	0	100	200	300
Pr. (Pascal)	4×10^3	3.5×10^3	3×10^3	2.5×10^3

Determine the order of reaction and its rate constant in Pa/sec.

CK0227

29. The half life period of decomposition of a compound is 50 minutes. If the initial concentration is halved, the half life period is reduced to 25 minutes. What is the order of reaction?

CK0029

30. For a chemical reaction A + B → products, the order is one with respect to each A and B. The sum of x and y from the following data is.

Rate (mol $l^{-1}s^{-1}$)	**[A](mol l^{-1})**	**[B](mol l^{-1})**
0.10	0.20	0.05
0.40	x	0.05
0.80	0.40	y

CK0030

CALCULATION OF RATE CONSTANT USING DIFFERENT PARAMETERS

31. For the Ist order reaction

$A \longrightarrow B + C$

Time	t	∞
Total pressure of (A + B + C)	P_2	P_3

If $k = \frac{1}{t}\ln\frac{P_3}{(xP_3 - yP_2)}$ then x + y will be. **CK0228**

32. For the Ist order reaction $S \longrightarrow G + F$

Time	t	∞
Rotation of (G + F)	r_t	r_∞

If $k = \frac{1}{t}\ln\left(\frac{xr_\infty}{r_\infty - zr_t}\right)$ then x + z will be **CK0229**

33. The thermal decomposition of dimethyl ether was measured by finding the increase in pressure at 500ºC in the reaction

$(CH_3)_2O(g) \longrightarrow CH_4(g) + H_2(g) + CO(g)$

Time (sec.)	200	400	∞
Change in Pressure (mm Hg)	540	594	600

The initial pressure of ether was 300 mm Hg. Determine the rate constant in $M^{1-n}sec^{-1}$ here is n is order of reaction. **CK0230**

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34. From the following data, calculate value of rate constant for decomposition of H_2O_2 in M^{-1} min^{-1}

Time (in minutes)	0	10
Volume (in c.c. of $KMnO_4$)	22.8	11.4

CK0231

35. The following data were obtained in experiment of inversion of cane sugar.

Time (minutes)	0	60	120	180	∞
Angle of rotation (degree)	+15	+ 7	+ 3	+1	–1

After what time (in min) would you expect a zero reading in polarimeter?

CK0232

36. The decomposition of N_2O_5 according to the equation $2\ N_2O_5\ (g) \longrightarrow 4\ NO_2(g) + O_2(g)$ is a first order reaction. After 60 min. from start of decomposition in a closed rigid vessel, the total pressure developed is found to be 350 mm Hg. On complete decomposition, the total pressure is 500 mm Hg. Calculate the rate constant for disappearance of $N_2O_5(k_{N_2O_5})$ in min^{-1} for the reaction.

CK0233

37. The reaction given below, rate constant for disappearance of A is 8×10^{-3} sec^{-1}. Calculate the time in second required for the total pressure in a system containing A at an initial pressure of 0.1 atm to rise to 0.145 atm and also find the total pressure in atm after 100 sec. [ln7 = 1.9 , ln 10 = 2.3].

$$2A\ (g) \longrightarrow 4B(g) + C(g)$$

CK0037

38. The reaction $A(aq) \longrightarrow B\ (aq) + C\ (aq)$ is monitered by measuring optical rotation of reaction mixture at different time interval. The species A, B and C are optically active with specific rotations 20°, 30° and – 40° per mole respectively. Starting with pure A, if the value of optical rotation was found to be 2.5° after 6.93 minutes and optical rotation was –5° after infinite time. Find the rate constant in min^{-1} for first order conversion of A into B and C. **CK0038**

39. For the hydrolysis of ethyl acetate in presence of H^+ as catalyst.

$$CH_3COOC_2H_5(aq.) + H_2O \xrightarrow{H^+} CH_3COOH(aq) + C_2H_5OH(aq.)$$

the experimentally observed rate law is

$$\text{rate} = k\ [\text{ester}]\ [H_2O]\ [H^+].$$

If the value of rate constant is 1.386×10^{-3} M^{-2} min^{-1} and concentration of $[H^+]$ ion is 1.8 M, how many seconds will it take for concentration of ester to become half of initial value ?

CK0234

40. If 0.01 % of a substance undergoing decomposition is consumed in 1 milli seconds when the concentration is 0.02M and in 0.25 milli seconds when the concentration is 0.04M. The order of reaction is.

CK0040

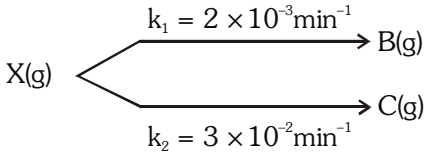
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PARALLEL AND REVERSIBLE REACTIONS

41. $A \xrightarrow{k_1} B$, $A \xrightarrow{k_2} C$, $k_1 = 1/11\ hr^{-1}$; $k_1 : k_2 = 1 : 10$. Calculate $\frac{[C]}{[A]}$ after one hour from the start of the reaction. Assuming only A was present in the beginning. [e = 2.7]

CK0235

42. A gaseous reactant **X** decompose to produce gaseous product B & C in a parallel reaction, both by first order, as follows :



If the decomposition is carried out in a sealed flask, partial pressure of B after very long time was found to be 100 mm Hg. Determine the time in minute when pressure of **X**(g) was 100 mm Hg. [ln2 = 0.693] **CK0236**

43. The reaction A proceeds in parallel channels $A \begin{matrix} \nearrow B \\ \searrow C \end{matrix}$. Suppose the half life in min. for the two branches are 60 minutes and 90 minutes, what is the overall half-life value in min.? **CK0044**

44. 3 mole mixture of two different substances A & B (mole fraction of A = 2/3) undergoes parallel first order reaction to form product 'C' as follow at 300 K.

$A \xrightarrow{k_1} C \qquad t_{1/2} = 2hr$

$B \xrightarrow{k_2} C \qquad t_{1/2} = 4hr$

The time in hr at which there will be 2 mole of 'C' is **CK0047**

45. For the series of competitive reactions :

$H + HO_2 \xrightarrow{K_1} H_2 + O_2$

$H + HO_2 \xrightarrow{K_2} 2OH$

$H + HO_2 \xrightarrow{K_3} H_2O + O$

It has been found that $K_1 : K_2 : K_3 = 0.60 : 0.30 : 0.10$. The molar ratio of the product, H_2 to H_2O at time t, is - **CK0048**

TEMPERATURE DEPENDENCE OF RATE

46. In gaseous reactions important for understanding the upper atmosphere, H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and E_a=77 kJ mol^{-1}, then calculate E_a in kJ/mole for the bimolecular recombination of 2OH radicals to form H_2O & O at 500 K

CK0051

47. The specific reaction rate for a reaction increases by a factor of 4, if the temperature is changed from 27°C to 47°C. Find the activation energy (in kcal) for the reaction (Use ln 2 = 0.7)

CK0052

48. Given that the temperature coefficient for the saponification of ethyl acetate by NaOH is 1.75. Calculate activation energy in kcal /mole for the saponification of ethyl acetate. Let initial temperature is 298K. [**Use :** ln (1.75) = 0.56] **CK0053**

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49. The energy of activation for a reaction at 27ºC is 10 kJ/mol. The presence of catalyst lowers the energy of activation by 75%. Find the factor by which rate of reaction increases at 27ºC due to catalyst.

(Take R = 25/3 J/mol-k) [e = 2.7]

CK0237

50. At 380°C, the half–life period for the first order decomposition of H_2O_2 is 360 min.The energy of activation of the reaction is 200 kJ mol^{-1}. Calculate the time required in minute for 75% decomposition at 450°C. **Use :** $\left[\frac{1}{653}-\frac{1}{723}\right]=1.5\times10^{-4}$, $e^{3.6}=36$, $R=\frac{25}{3}J/k-mol$

CK0238

51. The rate constant for decomposition of $COCl_2$ (g) according to following reaction

$COCl_2(g) \rightarrow CO(g) + Cl_2(g)$; ΔH = 19 kcal/mol, is expressed as

$\ln k = 15 - \frac{5000}{T}$

Calculate activation energy for given reaction (in Kcal/mol.) **CK0056**

52. At 300 K, 50% of molecule collide with energy greater than or equal to E_a. At what temperature in K, 25% molecule will have energy greater than or equal to E_a. **CK0057**

53. For a zero order reaction at 200K reaction completes in 5 minutes while at 300K, same reaction completes in 2.5 minutes. What will be the activation energy in calorie.

(R = 2 Cal/mol-k; *l*n 2 = 0.7) **CK0058**

54. For a first order reaction : A → P, the temperature (T) dependent rate constant (k) was found to follow the equation $\log_{10}k=-(2000)\frac{1}{T}+6$. Then activation energy (in kJ/mole) equation of reaction will be. (ln x = 2.3 × log x)

CK0239

55. A reaction takes place in three steps. The rate constants are k_1 , k_2 , k_3. The overall rate constant $k=\frac{k_1\sqrt{k_3}}{k_2}$. If energy of activation is 40 , 30 and 20 kJ /mole respectively, the overall energy of activation in kJ/mole is :

CK0240

MECHANISM OF REACTION

56. The reaction $2NO + Br_2 \longrightarrow 2NOBr$, is supposed to follow the following mechanism

(i) $NO + Br_2 \overset{\text{fast}}{\rightleftharpoons} NOBr_2$ (ii) $NOBr_2 + NO \xrightarrow{\text{slow}} 2NOBr$

total order of reaction will be

CK0241

57. For the reaction $2H_2 + 2NO \longrightarrow N_2 + 2H_2O$, the following mechanism has been suggested :

$2NO \rightleftharpoons N_2O_2$ equilibrium constant = K_1 (fast)

$N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$ (slow)

$NO + H_2 \xrightarrow{k_3} N_2 + H_2O$ (fast)

Order of reaction w.r.t. NO will be **CK0242**

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58. For the reaction :

$$2O_3(g) \longrightarrow 3O_2(g)$$

Mechanism :

Step - I $O_3(g) \underset{}{\overset{\text{fast}}{\rightleftharpoons}} O_2(g) + O(g)$

Step - II $O(g) + O_3(g) \xrightarrow{\text{slow}} 2O_2(g)$

Overall order of reaction based on mechanism is **CK0063**

59. A complex reaction : $2X + Y \rightarrow Z$, takes place in two steps :

$X + Y \xrightarrow{K_1} 2W$ **CK0064**

$X + 2W \xrightarrow{K_2} Z$

If $K_1 << K_2$, order of reaction will be -

60. The suggested mechanism for the reaction : $CHCl_3(g) + Cl_2(g) \rightarrow CCl_4(g) + HCl(g)$, is -

$Cl_2 \underset{K_2}{\overset{K_1}{\rightleftharpoons}} 2\dot{C}l$(fast)

$CHCl_3 + \dot{C}l \xrightarrow{K_3} HCl + \dot{C}Cl_3$ (slow)

$\dot{C}Cl_3 + \dot{C}l \xrightarrow{K_4} CCl_4$(fast)

Total order of reaction will be

CK0243

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EXERCISE # S-II

1. The decomposition of a compound P, at temperature T according to the equation

$$2P_{(g)} \longrightarrow 4Q_{(g)} + R_{(g)} + S_{(l)}$$

is the first order reaction. After 30 minutes from the start of decomposition in a closed vessel, the total pressure developed is found to be 317 mm Hg and after a long period of time the total pressure observed to be 617 mm Hg. Calculate the total pressure (in mm Hg) of the vessel after 75 minutes, if volume of liquid S is supposed to be negligible. Also calculate the time fraction $t_{7/8}$ (in minutes). [Give your answers as nearest integers]

Given : Vapour pressure of S (l) at temperature T = 32.5 mm Hg.

[ln = 1.169) = 0.156, ln2 = 0.7, $e^{0.39} = 1.5$]

CK0066

2. For the reaction : A + B $\longrightarrow$ Product, rate law is : rate = k [A]^2 [B], where k = 5×10^{-5} $(mol/L)^{-2}$ min^{-1} . Determine the **time (in minutes)** in which concentration of 'A' becomes half of its initial concentration, if initial concentration of A and B are 0.2 M and 2×10^3 M respectively.

CK0067

3. An optically active compound A upon acid catalysed hydrolysis yield two optically active compound B and C by pseudo first order kinetics. The observed rotation of the mixture after 20 min was 5° while after completion of the reaction it was – 20°. If optical rotation per mole of A, B & C are 60°, 40° & – 80°. Calculate half life of the reaction (in minutes).

CK0068

4. For the two parallel first order reactions A $\xrightarrow{k_1}$ B and A $\xrightarrow{k_2}$ C, show that the activation energy E′ for the disappearance of A is given in terms of activation energies E_1 and E_2 for the two paths by

$$E' = \frac{k_1E_1 + k_2E_2}{k_1 + k_2}$$

CK0070

5. At room temperature (27°C), orange juice gets spoilt in about 64 hours. In a refrigerator at 7°C, juice can be stored three times as long before it gets spoilt. Estimate

(a) the activation energy of the reaction that causes the spoiling of juice (in kcal/mole).

(b) How long should it take (in hours) for juice to get spoilt at 47°C?

[$e^{0.9625} = 2.5$, ln3 = 1.1 , R = 2 cal/mol-K]

CK0071

6. Two reactions (i) A $\rightarrow$ products (ii) B $\rightarrow$ products, follow first order kinetics. The rate of the reaction (i) is doubled when the temperature is raised from 300 K to 310K. The half life for this reaction at 310K is 30 minutes. At 310K temperature and same concentration of reactant in both reaciton B, decomposes twice as fast as A. If the energy of activation for the reaction (ii) is half that of reaction (i), calculate the rate constant of the reaction (ii) at 300K (in min^{-1}).

CK0072

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7. The reaction of formation of phosgene from CO and Cl_2 is $CO + Cl_2 \longrightarrow COCl_2$
The proposed mechanism is

(i) $Cl_2 \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} 2Cl$ (fast equilibrium)

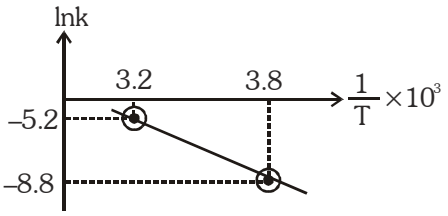
(ii) $Cl + CO \underset{K_{-2}}{\overset{K_2}{\rightleftharpoons}} COCl$ (fast equilibrium)

(iii) $COCl + Cl_2 \xrightarrow{K_3} COCl_2 + Cl$ (slow)

Show that the above mechanism leads to the following rate law $\frac{d[COCl_2]}{dt} = K[CO][Cl_2]^{3/2}$,

where $K = k_3 . \frac{k_2}{k_{-2}} \left(\frac{k_1}{k_{-1}}\right)^{1/2}$. **CK0073**

8. For an uncatalyzed elementary reaction, following observation is made :



If the above reaction is carried out in the presence of catalyst at 300K, the reaction proceeds with same rate as without catalyst at 400K. By what amount (in kcal) catalyst has decreased E_a.
CK0074

9. Milk is pasteurised if it is heated at 67ºC for 4 hours. If $\frac{E_a}{2.303}$ for the process is 23.8 kcal / mol then minimum how much seconds will be required for the the process at 77ºC ? [R = 2cal /K-mol]
CK0075

10. When CrI_3 reacts with H_2O_2 in presence of NaOH forming Na_2CrO_4, $NaIO_4$ & H_2O then find how many times will be rate of disappearance of NaOH compared to CrI_3. **CK0076**

11. Conversion of A into B and C follows first order kinetics. A, B and C all are optically active, initially A is present in 500 ml solution and 100 minutes after the start of reaction, the resulting solution becomes optically inactive. If angle of rotation of A is +50 degree per molar, B is –20 degree per molar and C is +10 degree per molar, then find the time for completion of $\left(\frac{875}{9}\right)\%$ of the chemical reaction in minutes.

A(aq.) → B(aq.) + C(aq.) **CK0077**

12. A gas phase reaction is,
$A(g) + 2B(g) \rightarrow C(g)$
The reaction was carried out with stoichiometric proportions of A and B and the following data is obtained

Half life (min)	**10**	**160**
Initial pressure of A(atm)	**40 atm**	**10 atm**

What is overall order of reaction ? **CK0078**

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13. For the reaction ,

$A \longrightarrow P$

If ratio of $t_{7/8} : t_{1/4}$ is 7 : 2 what is order of reaction.

$t_{7/8}$ = time in which 7/8 part of reactant is reacted

$t_{1/4}$ = time in which 1/4 part of reactant is reacted **CK0079**

14. A certain reactant A^{n+} is getting converted to $A^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing reagent which only reacts with A^{n+} and $A^{(n+4)+}$. In this process, it converts A^{n+} to $A^{(n-2)+}$ and $A^{(n+4)+}$ to $A^{(n-1)+}$.

Time	0	10 min
Volume of reagent consumed	30 ml	45 ml

Calculate the rate constant (in min^{-1}) of the conversion of A^{n+} to $A^{(n+4)+}$ assuming it to be first order reaction.

[n-factor of reagent remain same when it reacts with A^{n+} & $A^{(n+4)+}$]

[Given: *l*n 3/2 = 0.4] **CK0080**

15. Consider two Ist order reactions at 25ºC with same initial concentrations of 1M

I. $A \xrightarrow{t_{1/2} = 40\,min}$ product (temperature coefficient = 2)

II. $B \xrightarrow{t_{1/2} = 30\,min}$ product (temperature coefficient = 3)

If both reactions are carried out at 35ºC. Find ratio of concentrations of A & B after one hour.

CK0081

16. Consider two reactions at 27ºC

(I). $A \longrightarrow B$; (First order kinetics)

(II). $X \longrightarrow Y$; (Zero order kinetics)

If both A & X each having initial concentration 0.1M & same half life period. Then find simplest ratio of rate constants for I & II reactions. [ln2 = 0.7] **CK0082**

17. The rate of first-order reaction is 0.04 mol $litre^{-1}s^{-1}$ at 10 minutes and 0.03 mol $litre^{-1}s^{-1}$ at 22 minutes after initiation. The half life of the reaction (in seconds) is (ln2 = 0.7, ln3 = 1 .1).

CK0083

18. In order to determine the order of reaction : $A(g) \rightarrow 2B(g)$, vapour density of the system is determined at different stages of reaction, at constant temperature. The reaction is started with pure A(g). From the following data, the order of reaction is -

Time (min)	0	10	20	
Vapour density	42	35	30	**CK0084**

19. In the reaction : $A \rightarrow P$, the rate is doubled when the concentration of 'A' is quadrupled. If 50% of the reaction occurs in $8\sqrt{2}$ hr , how long (in hours) would it take for the completion of next 50% reaction. **CK0085**

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EXERCISE # O-I

SINGLE CORRECT

RATE OF REACTION

1. The rate of a reaction is expressed in different ways as follows :

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is:

(A) $4A + B \longrightarrow 2C + 3D$ (B) $B + 3D \longrightarrow 4A + 2C$

(C) $A + B \longrightarrow C + D$ (D) $B + D \longrightarrow A + C$ **CK0086**

2. For the reaction $2A + 3B \rightarrow 4C$

the rate of reaction may be represented as :–

(A) $r = -2\frac{d(A)}{dt} = -3\frac{d(B)}{dt} = 4\frac{d(C)}{dt}$ (B) $r = -2\frac{d(A)}{dt} = -4\frac{d(B)}{dt} = 3\frac{d(C)}{dt}$

(C) $r = -\frac{1}{2}\frac{d(A)}{dt} = \frac{1}{3}\frac{d(B)}{dt} = \frac{1}{4}\frac{d(C)}{dt}$ (D) $r = -\frac{1}{2}\frac{d(A)}{dt} = -\frac{1}{3}\frac{d(B)}{dt} = \frac{1}{4}\frac{d(C)}{dt}$

CK0087

3. In a reaction $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$, the rate of appearance of NH_3 is 2.5×10^{-4} molL^{-1}sec^{-1}. The rate of reaction & rate of disappearance of H_2 will be (In molL^{-1}sec^{-1})

(A) 3.75×10^{-4}, 1.25×10^{-4} (B) 1.25×10^{-4}, 2.5×10^{-4}

(C) 1.25×10^{-4}, 3.75×10^{-4} (D) 5.0×10^{-4}, 3.75×10^{-4} **CK0088**

4. For the reaction $4A + B \rightarrow 2C + 2D$, the incorrect statement is :–

(A) The rate of disappearance of B is one fourth the rate of disappearance of A

(B) The rate of appearance of C is half the rate of disappearance of B

(C) The rate of formation of D is half the rate of consumption of A

(D) The rates of formation of C and D are equal **CK0089**

5. Which of the following rate law has an overall order of 0.5 for reaction involving substances x,y and z ?

(A) Rate = $k [C_x] [C_y] [C_z]$ (B) Rate = $k [C_x]^{0.5}[C_y]^{0.5}[C_z]^{0.5}$

(C) Rate = $k [C_x]^{1.5} [C_y]^{-1}[C_z]^{\circ}$ (D) Rate = $k [C_x][C_z]^{\circ} / [C_y]^2$ **CK0090**

6. For a reaction A + B $\rightarrow$ products, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the conc. of A & B were doubled. The order of the reaction with respect to A & B are:–

(A) 1, 1 (B) 2, 0 (C) 1, 0 (D) 0, 1 **CK0091**

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7. If the rate of the reaction is equal to the rate constant, the order of the reaction is:-

(A) 0 (B) 1 (C) 2 (D) 3 **CK0092**

8. If the first order reaction involves gaseous reactants and gaseous products the units of its rate may be –

(A) atm. (B) atm - sec. (C) atm – sec^{-1} (D) atm^2 sec^2

CK0093

9. If concentration of reactants is increased by 'x', then the rate constant (k) becomes –

(A) $\ln\frac{k}{x}$ (B) $\frac{k}{x}$ (C) k + x (D) k **CK0094**

10. The rate constant of nth order reaction has units

(A) $litre^{1-n}$ mol^{1-n} sec^{-1} (B) mol^{n-1} $litre^{1-n}$ sec^{-1}

(C) mol^{1-n} $litre^{n-1}$ sec^{-1} (D) None **CK0095**

11. A reaction is found to have the rate constant x sec^{-1}. By what factor the rate is increased if initial conc. of A is tripled ?

(A) 3 (B) 9 (C) x (D) Remains same

CK0096

12. Consider following two reactions

$A \longrightarrow$ Product ; $-\frac{d[A]}{dt} = k_1 [A]^0$

$B \longrightarrow$ Product ; $-\frac{d[B]}{dt} = k_2 [B]$

k_1 and k_2 are expressed in terms of molarity (mol L^{-1}) and time (sec) as

(A) sec^{-1}, M sec^{-1} (B) M sec^{-1}, M sec^{-1}

(C) sec^{-1}, M^{-1} sec^{-1} (D) M sec^{-1}, sec^{-1} **CK0097**

13. $A(g) \longrightarrow B(g) + 3C(g)$

In a closed container at a given temperature, pressure increases from 100 mm Hg to 160 mm Hg in 10 sec. for reaction.

Then the average rate of reaction in first 10 sec. will be -

(A) 2 mm/sec. (B) 4 mm/sec. (C) 6 mm/sec. (D) 3 mm/sec.

CK0098

ZERO ORDER REACTION

14. H_2 gas is adsorbed on the metal surface like tungsten. This follows ..... order reaction -

(A) Third (B) Second (C) Zero (D) First **CK099**

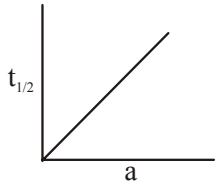
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15. The rate constant of a zero order reaction is 0.2 mol $dm^{-3}h^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3}. Then its initial concentration would be :

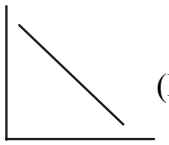
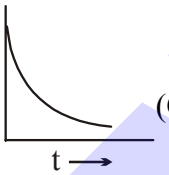
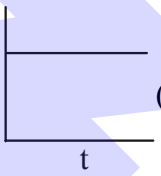
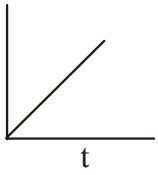
(A) 0.15 mol dm^{-3} (B) 1.05 mol dm^{-3}

(C) 0.25 mol dm^{-3} (D) 4.00 mol dm^{-3} **CK0100**

16. Consider the reaction A $\longrightarrow$ B, graph between half life ($t_{1/2}$) and initial concentration (a) of the reactant is

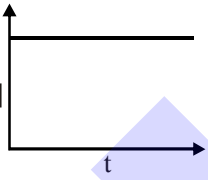
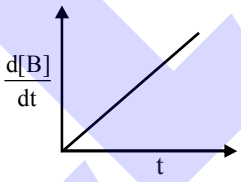
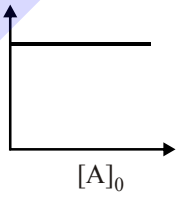
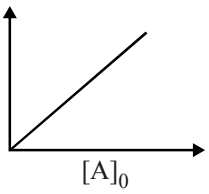


Hence graph between $-\frac{d[A]}{dt}$ and time will be

(A)  (B)  (C)  (D) 

CK0101

17. Which graph represents zero order reaction [A(g) → B(g)] :

(A)  (B)  (C)  (D) 

CK0102

18. K for a zero order reaction is 2×10^{-2} $molL^{-1}$ sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M, then concentration after 50 sec. must have been.

(A) 0.5 M (B) 0.25 M (C) 0.125 M (D) 0.0 M **CK0103**

FIRST ORDER REACTION

19. The rate constant of a first order reaction is 4×10^{-3} sec^{-1}. At a reactant concentration of 0.02 M, the rate of reaction would be –

(A) 8×10^{-5} M sec^{-1} (B) 4×10^{-3} M sec^{-1}

(C) 2×10^{-1} M sec^{-1} (D) 4×10^{-1} M sec^{-1} **CK0104**

20. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be –

(A) 10 min^{-1} (B) 6.931 min^{-1} (C) 0.6931 min^{-1} (D) 0.06931 min^{-1}

CK0105

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21. A first order reaction has a half life period of 69.3 sec. At 0.10 mol lit^{-1} reactant concentration rate will be –

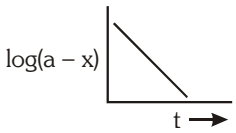
(A) 10^{-4} M sec^{-1} (B) 10^{-3} M sec^{-1} (C) 10^{-1} M sec^{-1} (D) 6.93×10^{-1} Msec^{-1}

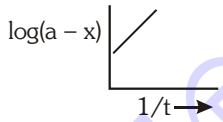
CK0106

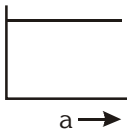
22. What fraction of a reactant (in first order reaction) is left after 40 minute if $t_{½}$ is 20 minute

(A) 1/4 (B) 1/2 (C) 1/8 (D) 1/6 **CK0107**

23. Which of the following curve represents a Ist order reaction :–

(A) 

(B) 

(C) 

(D) 1 & 3 both **CK0108**

24. After how many seconds will the conc. of the reactant in a first order reaction be halved, if the rate constant is 1.155×10^{-3} sec^{-1} :-

(A) 600 (B) 100 (C) 60 (D) 10 **CK0109**

25. Correct statement about first order reaction is:-

(A) $t_{completion}$ = finite (B) $t_{1/2} \propto \frac{1}{a}$

(C) Unit of k is mole lit^{-1} sec^{-1} (D) $t_{1/2} \times k$ = const. **CK0110**

26. For a given reaction of first order, it takes 20 minute for the concentration to drop from 1 M to 0.6 M. The time required for the concentration to drop from 0.6 M to 0.36 M will be :

(A) More than 20 min (B) Less than 20 min

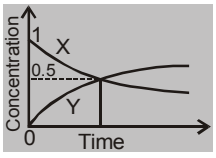
(C) Equal to 20 min (D) Infinity **CK0111**

27. The accompanying figure depicts the change in concentration of species X and Y for the reaction $X \rightarrow Y$ as a function of time. The point of intersection of the two curves represents.

(A) $t_{1/2}$

(B) $t_{3/4}$

(C) $t_{2/3}$



(D) Data insufficient to predict **CK0112**

28. A reaction is of first order. After 100 minutes, 75 g of the reactant A are decomposed when 100g are taken initially. Calculate the time required when 150g of the reactant A are decomposed, the initial weight taken is 200g.

(A) 100 minutes (B) 200 minutes (C) 150 minutes (D) 175 minutes

CK0113

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29. Consider the reaction :

$A \longrightarrow B + C$

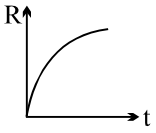
Initial concentration of A is 1 M. 20 minutes time is required for completion of 80 % reaction.

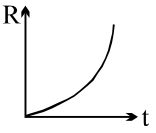
If $\frac{d[B]}{dt} = k[A]$, then half life ($t_{1/2}$) is (Use : ln 5 = 1.6, ln2 = 0.7)

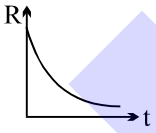
(A) 55.44 min. (B) 50 min (C) 8.75 min (D) 12.5 min

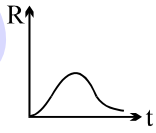
CK0114

30. If decomposition reaction A (g) $\longrightarrow$ B (g) follows first order kinetics, then the graph of rate of formation (R) of B against time t will be

(A) 

(B) 

(C) 

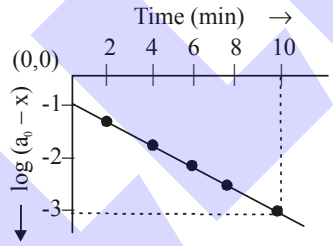
(D) 

CK0115

31. For the first order decomposition of $SO_2Cl_2(g)$,

$SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$

a graph of log $(a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?



(A) 0.2 (B) 4.6×10^{-1} (C) 7.7×10^{-3} (D) 1.15×10^{-2}

CK0116

32. The rate constant for a second order reaction is 8×10^{-5} M^{-1} min^{-1}. How long will it take a 1M solution to be reduced to 0.5 M ?

(A) 8.665×10^{3} min (B) 8×10^{-3} min (C) 1.25×10^{4} min (D) 4×10^{-5} min

CK0117

33. The rate law of the reaction : A + 2B $\rightarrow$ product(P) is given by $\frac{d[P]}{dt} = K[A]^2[B]$. If A is taken in large excess, the order of the reaction will be –

(A) Zero (B) 1 (C) 2 (D) 3 **CK0118**

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CALCULATION OF ORDER OF REACTION

34. Time required to complete a half fraction of a reaction varies inversely to the concentration of reactant then the order of reaction is –

(A) Zero (B) 1 (C) 2 (D) 3 **CK0119**

35. The reaction L → M is started with 10 g of L. After 30 and 90 minute, 5 g and 1.25 g of L are left respectively. The order of reaction is –

(A) 0 (B) 2 (C) 1 (D) 3 **CK0120**

36. From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, [$t_{1/2} \propto a$] is found. The order of reaction is :–

(A) 0 (B) 1 (C) 2 (D) 3 **CK0121**

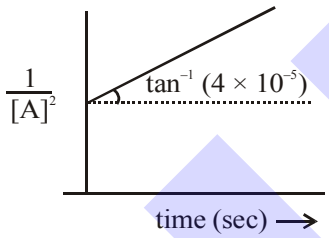
37. At certain temperature, the half life period for the thermal decomposition of a gaseous substance depends on the initial partial pressure of the substance as follows

P(mmHg)	500	250
$t_{1/2}$ (in min.)	235	950

Find the order of reaction [Given log (23.5) = 1.37 ; log (95) = 1.97; log 2 = 0.30]

(A) 1 (B) 2 (C) 2.5 (D) 3 **CK0122**

38. For a reaction A ⟶ Product



What is the value of k for the given reaction-

(A) $4 \times 10^{-5}\ M^{-1}s^{-1}$ (B) $\frac{4}{3} \times 10^{-5}\ M^{-2}s^{-1}$ (C) $2 \times 10^{-5}\ M^{-2}s^{-1}$ (D) $\frac{2}{3} \times 10^{-5}\ M^{-2}s^{-1}$

CK0123

39. Azo isopropane decomposes according to the equation :–

$$(CH_3)_2CHN{=}NCH(CH_3)_2(g) \xrightarrow{250-290^0\,C} N_2(g)+C_6H_{14}(g)$$

It is found to be a first order reaction. If initial pressure is P_o and pressure of the mixture at time t is (P_t) then rate constant K would be :–

(A) $k = \frac{2.303}{t} \log\frac{P_o}{2P_o - P_t}$ (B) $k = \frac{2.303}{t} \log\frac{P_o - P_t}{P_o}$

(C) $k = \frac{2.303}{t} \log\frac{P_o}{P_o - P_t}$ (D) $k = \frac{2.303}{t} \log\frac{2P_o}{2P_o - P_t}$ **CK0124**

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40. For a 1st order homogeneous gaseous reaction

$A \rightarrow 2B + C$,

if the pressure after time t was P_t and after long time was P_∞, then rate constant (k) in terms of P_t & P_∞ and t is -

(A) $k = \frac{2.303}{t}\log\left(\frac{P_\infty}{P_\infty - P_t}\right)$ (B) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{P_\infty - P_t}\right)$

(C) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{3(P_\infty - P_t)}\right)$ (D) None of these **CK0125**

41. The first order reaction

$A(g) \rightarrow B(g) + C(g) + D(s)$

taking place at constant pressure and temperature condition. Initially volume of the container containing only A was found to be V_0 and after time 't' it was V_t. Rate constant for the reaction is.

(A) $\frac{1}{t} ln \frac{V_0}{2V_0 - V_t}$ (B) $\frac{1}{t} ln \frac{V_0}{V_0 - V_t}$ (C) $\frac{1}{t} ln \frac{2V_0}{2V_0 - V_t}$ (D) $\frac{1}{t} ln \frac{2V_0}{V_0 + V_t}$

CK0126

42. For the inversion of cane sugar ($C_{12}H_{22}O_{11}$) obeying I order following data were obtained

Time (min.)	0	10	∞
Angle of rotation of solution (degree)	+20	–2.5	–10

What will be rate constant in min^{-1} (***ln* 2 = 0.7**)

(A) 0.7 (B) 0.14 (C) 0.21 (D) 0.07 **CK0127**

COLLISION THEORY AND ARRHENIUS EQUATION

43. The rate constant for the forward reaction A (g) $\rightleftharpoons$ 2B(g) is 1.5×10^{-3} s^{-1} at 100 K. If 10^{-5} moles of A and 100 moles of B are present in a 10 litre vessel at equilibrium then rate constant for the backward reaction at this temperature is

(A) 1.50×10^4 L mol^{-1} s^{-1} (B) 1.5×10^{11} L mol^{-1} s^{-1}

(C) 1.5×10^{10} L mol^{-1} s^{-1} (D) 1.5×10^{-11}L mol^{-1} s^{-1} **CK0128**

44. According to collision theory of reaction –

(A) Every collision between reactant leads to chemical reaction

(B) Rate of reaction is proportional to velocity of molecules

(C) All reactions which occur in gaseous phase are zero order reaction

(D) Rate of reaction is directly proportional to collision frequency. **CK0129**

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45. Activation energy of a reaction is –

(A) The energy released during the reaction

(B) The energy evolved when activated complex is formed

(C) Minimum amount of energy needed to overcome the potential barrier of reaction

(D) The energy needed to form one mole of the product **CK0130**

46. The minimum energy for molecules to enter into chemical reaction is called

(A) Kinetic energy (B) Potential energy

(C) Threshold energy (D) Activation energy **CK0131**

47. For producing the effective collisions, the colliding molecules must possess:-

(A) A certain minimum amount of energy

(B) Energy equal to or greater than threshold energy

(C) Proper orientation

(D) Threshold energy as well as proper orientation of collision **CK0132**

48. A large increase in the rate of a reaction for a rise in temperature is due to –

(A) Increase in the number of collisions (B) Increase in the number of activated molecules

(C) Lowering of activation energy (D) Shortening of the mean free path

CK0133

49. Slope of which plot can give the value of activation energy

(A) k versus T (B) $\frac{1}{k}$ versus T (C) Log k versus 1/T (D) C versus T

CK0134

50. Given that K is the rate constant for any order reaction at temperature T, then the value of $\lim_{T\to\infty} \log k$ _____.

(A) $\frac{A}{2.303}$ (B) A (C) 2.303 A (D) log A **CK0135**

51. For a certain gaseous reaction a 10° C rise of temperature from 25° C to 35° C doubles the rate of reaction. What is the value of activation energy :–

(A) $\frac{10}{2.303R \times 298 \times 308}$ (B) $\frac{2.303 \times 10}{298 \times 308R}$

(C) $\frac{0.693R \times 10}{290 \times 308}$ (D) $\frac{0.693R \times 298 \times 308}{10}$ **CK0136**

52. From the following data, the activation energy (cal/mol) for the reaction $H_2 + I_2 \rightarrow 2HI$, is about

T (in k)	**1/T, (in, k^{-1})**	**$\log_{10}k$**
769	1.3×10^{-3}	2.9
667	1.5×10^{-3}	1.1

(A) 4×10^4 (B) 2×10^4 (C) 8×10^4 (D) 3×10^4 **CK0137**

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53. The rate constant, the activation energy and the Arrhenius parameter (A) of a chemical reaction at 25°C are 3.0×10^{-4} s^{-1}, 104.4 kJ mol^{-1} and $6.0 \times 10^{14} s^{-1}$ respectively. The value of the rate constant at $T \to \infty$ is

(A) 2.0×10^{18} s^{-1} (B) 6.0×10^{14} s^{-1} (C) infinity (D) 3.6×10^{30} s^{-1}

CK0138

54. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 min at 47°C. The energy of activation of the reaction is

(A) 100 kJ/mol (B) 55.14 kJ/mol (C) 11.97 kJ/mol (D) 6.65 kJ/mol

CK0139

55. For the first order reaction A $\longrightarrow$ B + C, carried out at 27 ºC if 3.8×10^{-16} % of the reactant molecules can overcome energy barrier, the E_a (activation energy) of the reaction is **[log 3.8 = 0.58, 2.303 × 8.314 × 17.42 = 333.33]**

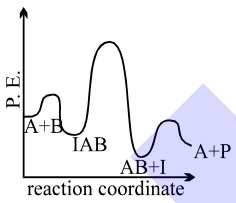
(A) 12 kJ/mole (B) 831.4 kJ/mole (C) 100 kJ/mole (D) 111.11 J/mole

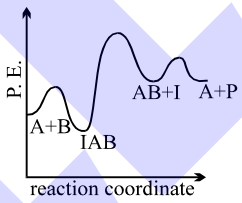
CK0140

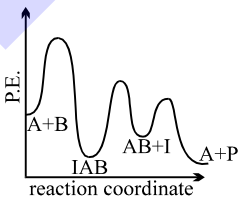
56. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

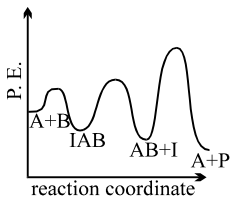
$$A + B \overset{\text{fast}}{\rightleftharpoons} I\,AB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

(A) 

(B) 

(C) 

(D) 

CK0141

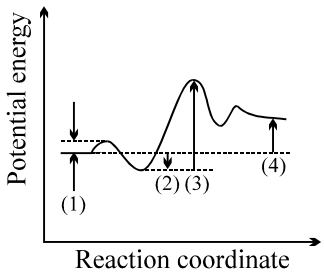
57. Choose the correct set of identifications for the reaction .

Substrate(S) $\xrightarrow{\text{Enzyme(E)}}$ Product (P)

whose mechanism is $E + S \rightleftharpoons ES$

$ES \rightleftharpoons EP$

$EP \rightleftharpoons E + P$



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	(1)	(2)	(3)	(4)
(A)	ΔE for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	$\Delta E_{overall}$ for $S \rightarrow P$	E_a for $EP \rightarrow E + P$
(B)	E_a for $E + S \rightarrow ES$	ΔE for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	$\Delta E_{overall}$ for $S \rightarrow P$
(C)	E_a for $ES \rightarrow EP$	E_a for $EP \rightarrow E + P$	$\Delta E_{overall}$ for $S \rightarrow P$	ΔE for $EP \rightarrow E + P$
(D)	E_a for $E + S \rightarrow ES$	E_a for $ES \rightarrow EP$	E_a for $EP \rightarrow E + P$	$\Delta E_{overall}$ for $S \rightarrow P$
(E)	ΔE for $E + S \rightarrow ES$	$\Delta E_{overall}$ for $S \rightarrow P$	ΔE for $_1$ $EP \rightarrow E + P$	E_a for $EP \rightarrow E + P$

CK0142

RATE LAW OF MECHANISM OF REACTION

58. Following mechanism has been proposed for a reaction : $2A + B \rightarrow D + E$

Step - I : $A + B \rightarrow C + D$ (slow)

Step - II : $A + C \rightarrow E$ (fast)

The rate law expression for the reaction is –

(A) $r = k[A]^2 [B]$ (B) $r = k[A] [B]$ (C) $r = k[A]^2$ (D) $r = k[A][C]$

CK0143

59. The reaction mechanism for the reaction $P \rightarrow R$ is as follows :

$$P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Q \text{ (fast)} \; ; \; 2Q + P \xrightarrow{K_3} R \text{ (slow)}$$

the rate law for the main reaction ($P \rightarrow R$) is :

(A) $k_1[P] [Q]$ (B) $k_1k_2[P]$ (C) $\frac{k_1k_3[P]^2}{k_2}$ (D) $k_1k_2[a]$

CK0144

60. The energies of activation for forward and reverse reactions for $A_2 + B_2 \rightleftharpoons 2AB$ are 180 kJ mol^{-1} and 200 kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol^{-1}. The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of catalyst will be (in kJ mol^{-1}) -

(A) 300 (B) 120 (C) 280 (D) –20 **CK0145**

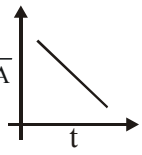
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EXERCISE # O-II

MORE THAN ONE MAY BE CORRECT

1. For the reaction A → B, the rate law expression is $-\frac{d[A]}{dt} = k\,[A]^{1/2}$. If initial concentration of [A] is $[A]_0$, then

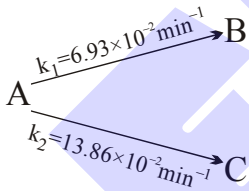
(A) The integerated rate expression is $k = \frac{2}{t}(A_0^{1/2} - A^{1/2})$

(B) The graph of $\sqrt{A}$ vs t will be 

(C) The half life period, $t_{1/2} = \frac{K}{2[A]_0^{1/2}}$

(D) The time taken for 75% completion of reaction $t_{3/4} = \frac{\sqrt{[A]_0}}{k}$ **CK0146**

2. Consider the reaction,



A, B and C all are optically active compound . If optical rotation per unit concentration of A, B and C are 60°, –72°, 42° and initial concentration of A is 2 M then select correct statement(s).

(A) Solution will be optically active and dextrorotatory after very long time

(B) Solution will be optically active and levorotatory after very long time

(C) Half life of reaction is 15 min

(D) After 75% conversion of A into B and C angle of rotation of solution will be 36°.

CK0147

3. Select **incorrect** statement(s):

(A) Unit of pre-exponential factor (A) for second order reaction is mol L^{-1} s^{-1}.

(B) A zero order reaction must be a complex reaction.

(C) Molecularity is defined only for RDS in a complex reaction.

(D) Rate constant (k) remain unaffected on changing temperature. **CK0148**

4. Which of the following is/are **correct** statement?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero order reaction, rate and the rate constant are identical.

(C) A zero order reaction is controlled by factors other than concentration of reactants.

(D) A zero order reaction is always elementary reaction. **CK0149**

5. For the gas phase reaction : R – H + X_2 → R – X + HX, following mechanism has been proposed

(i) $X_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2X^{\bullet}$ (ii) $X^{\bullet} + R-H \xrightarrow{k_3} R^{\bullet} + H-X$ (slowest)

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(iii) $R^{\bullet} + X_2 \xrightarrow{k_4} R-X + X^{\bullet}$

Based on this, select the correct option (s) :-

(A) Effective rate constant for the formation of RX is $k_3 k_4 \sqrt{\frac{k_1}{k_2}}$

(B) $\frac{d[RX]}{dt} \propto [X_2]$

(C) Overall order of the reaction is 3/2

(D) $\frac{d[RX]}{dt} \propto [RH]^1$ **CK0150**

6. For a first order reaction : $A(g) \rightarrow 2B(g)$

Time(in second)	0	20	40	∞
Total pressure of system (in mm.of Hg)	64	112	124	128

(A) Half life of reaction is 10 sec

(B) Value of rate constant for reaction is 6.93×10^{-3} sec^{-1}

(C) Total pressure at t = 50 sec will be 126 mm of Hg

(D) Reaction must be a complex reaction **CK0151**

7. Which of the following is **INCORRECT** for first order reaction ?

(A) On introducing catalyst, both rate constant and rate of reaction increases.

(B) On increasing temperature both rate constant & rate of reaction increases.

(C) On decreasing volume both rate constant & rate of gaseous reaction increases.

(D) On increasing concentration of gaseous reactant at constant volume & constant temperature both total pressure and rate of the reaction increases. **CK0152**

8. $2X(g) + Y(g) + 3Z(g) \rightarrow$ Products. The rate equation of above reaction is given by :

Rate = $K[X]^1[Y]^0[Z]^2$.

Choose the correct statements

(A) If [z] >> [x] and 75% of X undergoes reaction in 20 sec, then 50% of X will react in 10 sec.

(B) Rate of reaction decreases by reducing the concentration of Y to half of the original value

(C) The half life of Z increases by increasing its concentration if [x] >> [z]

(D) On doubling the concentration of each of X,Y & Z, rate of reaction becomes 8 times **CK0153**

9. Select the correct statement -

(A) In a mixture of $KMnO_4$ & $H_2C_2O_4$, $KMnO_4$ decolorises faster at higher temperature than lower temperature

(B) A catalyst participates in a chemical reaction by forming temporary bonds with the reactants resulting in an activated complex

(C) In collision theory only activation energy determine the criteria for effective collision

(D) Collision theory assumes molecules to be soft spheres & consider their structural aspects. **CK0154**

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10. For the reaction A + 3B → C, select the correct statement (s)-

(A) $\frac{d[C]}{dt} = \frac{-d[A]}{dt}$

(B) $\frac{3d[C]}{dt} = -\frac{d[B]}{dt}$

(C) Rate law must be r = k [A][B]^3

(D) Units for rate of reaction are independent of order of reaction **CK0155**

11. Choose the incorrect statement(s) -

(A) Activation energy of reaction always decreases on decreasing temperature

(B) Order of reaction may change with change in temperature

(C) When slowest step is the first step in a mechanism, then the rate law of overall reaction is the same as the rate law for this step

(D) Rate of photochemical reaction is directly proportional to intensity of absorbed photons.

CK0156

12. The forward rate constant of a reaction increases by 7% when its temperature is raised from 300K to 301K while its equilibrium constant increases by 3%. Which of the following is/are correct -

(A) Activation energy of forward reaction $(Ea)_F$ is $= 300 \times 301 \times R \times \ln(1.03)$

(B) Standard heat of reaction (ΔH°) is $= 300 \times 301 \times R \times \ln(1.07)$

(C) Activation energy of backward reaction $(Ea)_B$ is $= 300 \times 301 \times R \times \ln\left(\frac{1.07}{1.03}\right)$

(D) Activation energy of backward reaction $(Ea)_B$ is $= 300 \times 301 \times R \times \ln\left(\frac{1.03}{1.07}\right)$

CK0157

13. Which statement(s) is/are false ?

(A) Every reaction has defined overall order

(B) All rate constant have same dimension.

(C) Every chemical species that appears in the rate law of reaction must be reactant or product in that reaction.

(D) Rate constant is never negative. **CK0158**

ASSERTION & REASONING TYPE QUESTIONS

14. **Statement-1** : A fractional order reaction must be a complex reaction.

Statement-2 : Fractional order of RDS equals to overall order of a complex reaction.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CK0159**

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15. **Statement-1** : The time of completion of reactions of type A → product (order <1) may be determined.

Statement-2 : Reactions with order ≥ 1 are either too slow or too fast and hence the time of completion can not be determined.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CK0160**

16. **Statement-1** : In a reversible endothermic reaction, (E_{act}) of forward reaction is higher than that of backward reaction

Statement-2 : The threshold energy of forward reaction is more than that of backward reaction

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CK0161**

17. **Statement-1** : A catalyst provides an alternative path to the reaction in which conversion of reactants into products takes place quickly

Statement-2 : The catalyst forms an activated complex of lower potential energy, with the reactants by which more number of molecules are able to cross the barrier per unit of time.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CK0162**

COMPREHENSION

Paragraph for Question Nos. 18 & 19

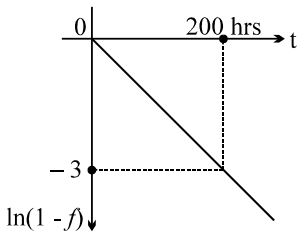
When metal surface is exposed to air if gets oxidised by first order kinetics if **(1–*f*)** is fraction of unoxidised. metal surface thickness and T is total thickness after very long time then it's rate can be represented as.

Rate law : $\frac{df}{dt}$ **= k(1 –** f **), where** f **= x/T,**

x = thickness of oxide film at time 't'

& T = thickness of oxide film at t = ∞

A graph of ln(1 – f) vs t is shown in the adjacent figure.



18. The time taken for thickness to grow 50% of 'T' is

(A) 23.1 hrs (B) 46.2 hrs (C) 100 hrs (D) 92.4 hrs **CK0163**

19. The exponential variation of 'f' with t(hrs) is given by

(A) $1 - e^{-3t/200}$ (B) $e^{-3t/200} - 1$ (C) $e^{-3t/200}$ (D) $e^{3t/200}$ **CK0163**

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Paragraph for Question Nos. 20 & 21

For a hypothetical elementary reaction : $A \xrightarrow{k_1} 2B$, $A \xrightarrow{k_2} 2C$ where $\frac{k_1}{k_2} = \frac{1}{2}$

Initially only 2 moles of A are present.

20. The total number of moles of A, B & C at the end of 50% reaction is -

(A) 2 (B) 3 (C) 4 (D) 5 **CK0164**

21. Number of moles of B at the end of 50% reaction , is -

(A) 2 (B) 1.333 (C) 0.667 (D) 0.333 **CK0164**

Paragraph for Question Nos. 22 & 23

A reaction is said to be first order if it's rate is proportional to the concentration of reactant. Let us consider a reaction

	$A(g)$	$\longrightarrow$	$B(g)$	+	$C(g)$
At t = 0	a		0		0
At time t	a – x		x		x

The rate of reaction is given by the expression $\frac{dx}{dt} = k(a-x)$ and integrated rate equation for a given reaction is represented as $k = \frac{1}{t} ln\left(\frac{a}{a-x}\right)$ where a = initial concentration and (a – x) = concentration of A after time t.

22. Thermal decomposition of compound X is a first order reaction. If 75% of X is decomposed in 100 min. How long will it take for 90% of the compound to decompose?

Given : log 2 = 0.30

(A) 190 min (B) 176.66 min (C) 166.66 min (D) 156.66 min

CK0165

23. Consider a reaction $A(g) \longrightarrow 3B(g) + 2C(g)$ with rate constant 1.386×10^{-2} min^{-1}. Starting with 2 moles of A in 12.5 litre a closed vessel initially, if reaction is allowed to takes place at constant pressure & at 298K then find the concentration of B after 100 min.

(A) 0.04 M (B) 0.36 M (C) 0.09 M (D) None of these

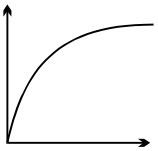
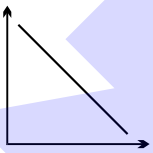
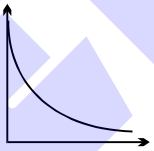
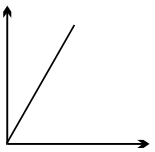
CK0166

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MATCH THE COLUMN

24. For the reaction of type $A(g) \longrightarrow 2B(g)$

Column-I contains four entries and **column-II** contains four entries. Entry of column-I are to be matched with **ONLY ONE ENTRY** of column-II

Column I	**Column II**
(A) $\frac{d[B]}{dt}$ vs $\frac{-d[A]}{dt}$ for first order	(P) 
(B) [A] vs t for first order	(Q) 
(C) [B] vs t for first order	(R) 
(D) [A] vs t for zero order	(S) 

CK0169

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25. Match the column :

	Column-I		**Column-II**
(A)	Inversion of cane sugar in excess water.	(P)	not 100% complete
(B)	saponification reaction with 1M NaOH	(Q)	pseudo-first order
(C)	decomposition of HI on gold	(R)	zero order
(D)	radioactive decay	(S)	second order

CK0170

26. For the reaction A + B → product, Given : $[A]_0 = [B]_0$

List-I (Observed Rate Law) is -

(P) $r = k[A]$

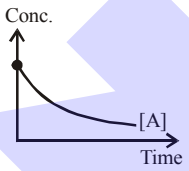
(Q) $r = k[A]^{1/2}[B]^{1/2}$

(R) $r = k[A]\,[B]$

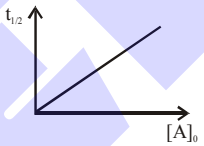
(S) $r = k[A]^0\,[B]^0$

List-II (Graph)

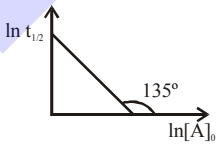
(1)



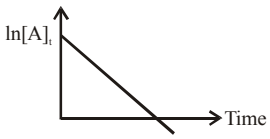
(2)



(3)



(4)



Code :

	P	**Q**	**R**	**S**
(A)	4	1	3	2
(B)	2	3	1	4
(C)	1	2	3	4
(D)	4	3	2	1

CK0170

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Match the Columns for Reaction A → P

Column - I	Column - II	Column - III
(I) First Order	(i) Reaction complete in finite time	(P) Rate depends on concentration
(II) Second Order	(ii) Reaction complete in infinite time	(Q) After equal interval of time concentration of reactant left are in G.P.
(III) Third Order	(iii) Half life is independent of concentration of reactant	(R) After equal interval of time concentration of reactant left are in A.P.
(IV) Zero Order	(iv) Half life decreases when concentration of reactant increases	(S) Half life depends on temperature

27. Select only incorrect option

(A) (II), (ii), (P) (B) (IV), (iii), (R)

(C) (III), (ii), (P) (D) (I), (ii), (Q) **CK0171**

28. Select only incorrect option

(A) (IV), (i), (S) (B) (III), (iv), (P)

(C) (I), (iv), (Q) (D) (II), (iv), (S) **CK0171**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Solid, Liquid & CK (PC)\Eng\03-Chemical kinetics\02_Ex.p65

EXERCISE # J-MAIN

1. Higher order (>3) reactions are rare due to :- **[JEE-MAIN-(Offline)2015]**

(1) shifting of equilibrium towards reactants due to elastic collision

(2) loss of active species on collision

(3) low probability of simultaneous collision of all the reacting species

(4) increase in entropy and activation energy as more molecules are involved. **CK0175**

2. Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :- **[JEE-MAIN-(Offline)2016]**

(1) 1.34×10^{-2} mol min^{-1} (2) 6.93×10^{-2} mol min^{-1}

(3) 6.93×10^{-4} mol min^{-1} (4) 2.66 L min^{-1} at STP **CK0178**

3. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below : **[JEE-MAIN-(Online)2016]**

$O_3(g) + Cl^{\cdot}(g) \rightarrow O_2(g) + ClO^{\cdot}(g)$ .....(i)

$k_i = 5.2 \times 10^9$ L $mol^{-1}s^{-1}$

$ClO^{\cdot}(g) + O^{\cdot}(g) \rightarrow O_2(g) + Cl^{\cdot}(g)$

$k_{ii} = 2.6 \times 10^{10}$ L $mol^{-1}s^{-1}$ .....(ii)

The closest rate constant for the overall reaction $O_3(g) + O^{\cdot}(g) \rightarrow 2O_2(g)$ is :

(1) 3.1×10^{10} L $mol^{-1}s^{-1}$ (2) 2.6×10^{10} L $mol^{-1}s^{-1}$

(3) 5.2×10^9 L $mol^{-1}s^{-1}$ (4) 1.4×10^{20} L $mol^{-1}s^{-1}$ **CK0179**

4. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to : ($R = 8.314$ J $mol^{-1}K^{-1}$) **[JEE-MAINS-2017]**

(1) 8 (2) 12 (3) 6 (4) 4 **CK0180**

5. The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A : **[MAINS-2017(online)]**

(1) 2.45 K (2) 4.92 K (3) 9.84 K (4) 19.67 K **CK0182**

6. At 518° C, the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is : **[JEE-MAINS-2018]**

(1) 3 (2) 1 (3) 0 (4) 2 **CK0183**

7. For a first order reaction, $A \rightarrow P$, $t_{1/2}$ (half life) is 10 days. The time required for $\frac{1}{4}^{th}$ conversion of A (in days) is :- ($\ln 2 = 0.693$, $\ln 3 = 1.1$) **[MAINS-2018(online)]**

(1) 5 (2) 4.1 (3) 3.2 (4) 2.5 **CK0184**

8. N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg. The pressure of the gaseous mixture after 100 minute at constant temperature will be: **[MAINS-2018(online)]**

(1) 116.25 mmHg (2) 175.0 mmHg (3) 106.25 mmHg (4) 136.25 mmHg **CK0185**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Solid, Liquid & CK (PC)\Eng\03-Chemical kinetics\02_Ex.p65

9. The following results were obtained during kinetic studies of the reaction :

$2A + B \rightarrow$ Products **[MAINS-2019(online)]**

Experiment	[A] (in mol L^{-1})	[B] (in mol L^{-1})	Initial Rate of reaction (in mol L^{-1} min^{-1})
(I)	0.10	0.20	6.93×10^{-3}
(II)	0.10	0.25	6.93×10^{-3}
(III)	0.20	0.30	1.386×10^{-2}

The time (in minutes) required to consume half of A is :

(1) 10 (2) 5 (3) 100 (4) 1 **CK0186**

10. For the reaction, $2A + B \rightarrow$ products, when the concentrations of A and B both wrere doubled, the rate of the reaction increased from 0.3 mol $L^{-1}s^{-1}$ to 2.4 mol L^{-1} s^{-1}. When the concentration of A alone is doubled, the rate increased from 0.3 mol $L^{-1}s^{-1}$ to 0.6 mol $L^{-1}s^{-1}$ **[MAINS-2019 (online)]**

Which one of the following statements is correct ?

(1) Order of the reaction with respect to B is 2 (2) Order of the reaction with respect to A is 2

(3) Total order of the reaction is 4 (4) Order of the reaction with respect to B is 1

CK0187

11. The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be :-

(1) 18.0 h (2) 7.2 h (3) 9.0 h (4) 12.0 h **CK0190**

12. If a reaction follows the Arrhenius equation, the plot lnk vs $\frac{1}{(RT)}$ gives straight line with a gradient (–y) unit. The energy required to activate the reactant is : **[MAINS-2019(online)]**

(1) y unit (2) –y unit (3) yR unit (4) y/R unit **CK0191**

13. Decomposition of X exhibits a rate constant of 0.05 µg/year. How many years are required for the decomposition of 5 µg of X into 2.5 µg ? **[MAINS-2019(online)]**

(1) 50 (2) 25 (3) 20 (4) 40

CK0193

14. It is true that : **[MAINS-2020(online)]**

(1) A zero order reaction is a single step reaction

(2) A second order reaction is always a multistep reaction

(3) A first order reaction is always a single step reaction

(4) A zero order reaction is a multistep reaction

CK0244

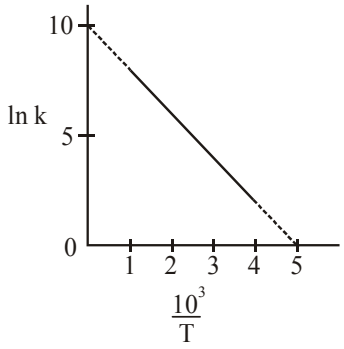
15. A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B(in s) : (Use ln 2 = 0.693) **[MAINS-2020(online)]**

(1) 180 (2) 120 (3) 300 (4) 900

CK0245

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16. The rate constant (k) of a reaction is measured at different temperatures (T), and the data are plotted in the given figure. The activation energy of the reaction in kJ mol^{-1} is : (R is gas constant)



[MAINS-2020(online)]

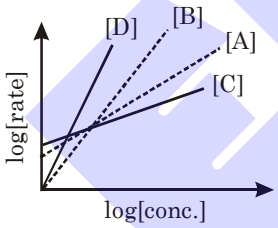
(1) 2R (2) R (3) 1/R (4) 2/R

CK0246

17. Consider the following reactions : **[MAINS-2020(online)]**

$A \rightarrow P1$; $B \rightarrow P2$; $C \rightarrow P3$; $D \rightarrow P4$

The order of the above reactions are a, b, c, and d, respectively. The following graph is obtained when log [rate] vs. log[conc] are plotted :

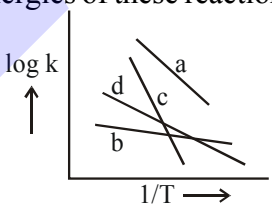


Among the following , the correct sequence for the order of the reactions is:

(1) a > b > c > d (2) c > a > b > d (3) d > b > a > c (4) d > a > b > c

CK0247

18. Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the activation energies of these reactions? **[MAINS-2020(online)]**



(1) $E_b > E_d > E_c > E_a$ (2) $E_a > E_c > E_d > E_b$

(3) $E_c > E_a > E_d > E_b$ (4) $E_b > E_a > E_d > E_c$

CK0248

19. For the reaction,

$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$

the observed rate expression is, rate = $k_f[NO]^2[H_2]$. The rate expression of the reverse reaction is :

(1) $k_b[N_2][H_2O]^2/[NO]$ (2) $k_b[N_2][H_2O]$

(3) $k_b[N_2][H_2O]^2$ (4) $k_b[N_2][H_2O]^2/[H_2]$

CK0249

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20. $2\ NO(g) + Cl_2(g) \rightarrow 2\ NOCl(s)$ **[MAINS-2021(online)]**

This reaction was studied at –10°C and the following data was obtained

run	$[NO]_0$	$[Cl_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall order of the reaction is _____. (Round off to the Nearest Integer).

CK0250

21. A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is _____ min. (Round off to the Nearest Integer). **[MAINS-2021(online)]**

CK0251

22. Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25°C. After 9 h, the fraction of sucrose remaining is *f*. The value of $\log_{10}\left(\frac{1}{f}\right)$ is _____ $\times 10^{-2}$. (Rounded off to the nearest integer) **[MAINS-2021(online)]**

[Assume : ln 10 = 2.303, ln 2 = 0.693]

CK0252

23. Gaseous cyclobutene isomerizes to butadiene in a first order process which has a 'k' value of $3.3 \times 10^{-4} s^{-1}$ at 153°C. The time in minutes it takes for the isomerization to proceed 40 % to completion at this temperature is _____. (Rounded off to the nearest integer) **[MAINS-2021(online)]**

CK0253

24. If the activation energy of a reaction is 80.9 kJ mol^{-1}, the fraction of molecules at 700 K, having enough energy to react to form products is e^{-x}. The value of *x* is _____. (Rounded off to the nearest integer) **[MAINS-2021(online)]**

[Use R = 8.31 J K^{-1} mol^{-1}]

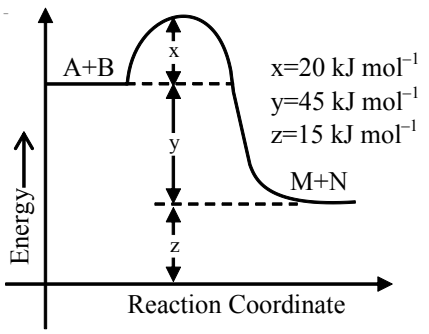
CK0254

25. The inactivation rate of a viral preparation is proportional to the amount of virus. In the first minute after preparation, 10% of the virus is inactivated. The rate constant for viral inactivation is _____ $\times 10^{-3}$ min^{-1}. (Nearest integer) **[MAINS-2021(online)]**

[Use : ln 10 = 2.303 ; $\log_{10}3$ = 0.477 ; property of logarithm : log x^y = y log x]

CK0255

26. According to the following figure, the magnitude of the enthalpy change of the reaction $A + B \rightarrow M + N$ in kJ mol^{-1} is equal to _____. (Integer answer) **[MAINS-2021(online)]**



CK0256

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EXERCISE # J-ADVANCED

1. For the first order reaction **[JEE 2011]**

$$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

(A) the concentration of the reactant decreases exponentially with time

(B) the half-life of the reaction decreases with increasing temperature.

(C) the half-life of the reaction depends on the initial concentration of the reactant.

(D) the reaction proceeds to 99.6% completion in eight half-life duration. **CK0194**

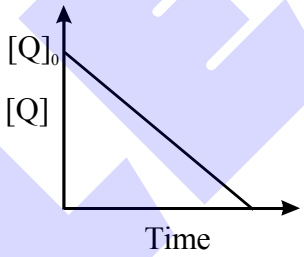
2. An organic compound undergoes first-order decomposition . The time taken for its decomposition to 1/8 and 1/10 of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{[t_{1/8}]}{t_{1/10}} \times 10$?

(take $\log_{10}2 = 0.3$) **[JEE 2012]**

CK0195

3. In the reaction : **[JEE 2013]**

$$P + Q \longrightarrow R + S$$



the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is -

(A) 2 (B) 3 (C) 0 (D) 1 **CK0196**

4. For the reaction **M** → **N**, the rate of disappearance of **M** increases by a factor of 8 upon doubling the concentration of **M**. The order of the reaction with respect to **M** is

(A) 4 (B) 3 (C) 2 (D) 1 **[JEE 2014]**

CK0197

5. In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate(II) is oxidized by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is. **[JEE 2015]**

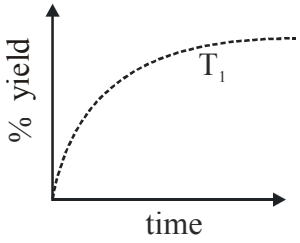
CK0198

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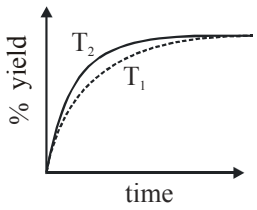
6. The % yield of ammonia as a function of time in the reaction **[JEE 2015]**

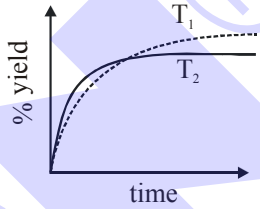
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H < 0$

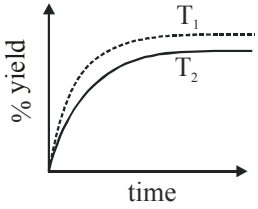
at (P , T_1) is given below -

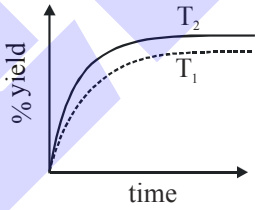


If this reaction is conducted at (P ,T_2), with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by -

(A) 

(B) 

(C) 

(D) 

CK0199

7. According to the Arrhenius equation, **[JEE 2016]**

(A) A high activation energy usually implies a fast reaction

(B) Rate constant increase with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy

(C) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant

(D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy. **CK0200**

8. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) : **[JEE 2017]**

(A) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

(B) The activation energy of the reaction is unaffected by the value of the steric factor

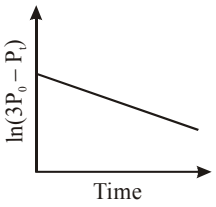
(C) Since P = 4.5, the reaction will not proceed unless an effective catalyst is used.

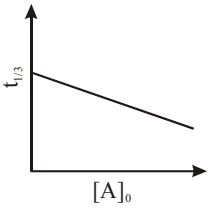
(D) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation. **CK0201**

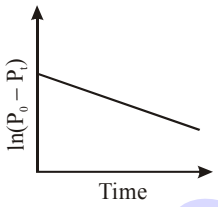
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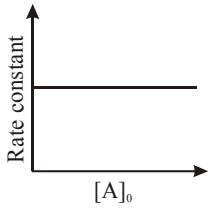
9. For a first order reaction $A(g) \rightarrow 2B(g) + C(g)$ at constant volume and 300 K, the total pressure at the beginning (t = 0) and at time t are P_0 and P_t, respectively. Initially, only A is present with concentration $[A]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach $1/3^{rd}$ of its initial value. The correct option(s) is (are) :- **[JEE 2018]**

(Assume that all these gases behave as ideal gases)

(A) 

(B) 

(C) 

(D) 

CK0202

10. Consider the kinetic data given in the following table for the reaction A + B + C → Product.

Experiment No.	[A] (mol dm^{-3})	[B] (mol dm^{-3})	[C] (mol dm^{-3})	Rate of reaction (mol dm^{-3} s^{-1})
1	0.2	0.1	0.1	6.0×10^{-5}
2	0.2	0.2	0.1	6.0×10^{-5}
3	0.2	0.1	0.2	1.2×10^{-4}
4	0.3	0.1	0.1	9.0×10^{-5}

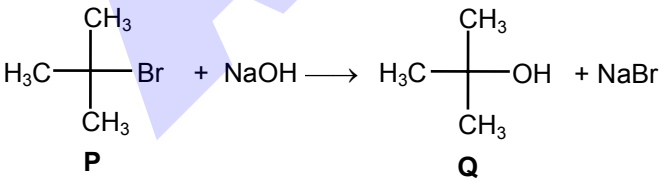
The rate of the reaction for [A] = 0.15 mol dm^{-3}, [B] = 0.25 mol dm^{-3} and [C] = 0.15 moldm^{-3} is found to be **Y** $\times 10^{-5}$ mol dm^{-3} s^{-1}. The value of **Y** is ———— . **[JEE 2019]**

CK0203

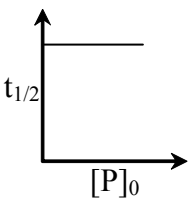
11. The decomposition reaction $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After Y $\times 10^3$ s, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is 5×10^{-4} s^{-1}, assuming ideal gas behaviour, the value of Y is ___ . **[JEE 2019]**

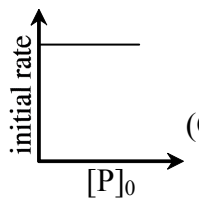
CK0204

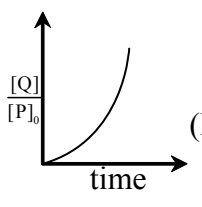
12. Which of the following plots is(are) correct for the given reaction (first order S_N1 reaction)? ($[P]_0$ is the initial concentration of P) **[JEE 2020]**

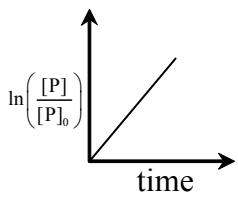


P **Q**

(A) 

(B) 

(C) 

(D) 

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13. For the following reaction **[JEE 2021]**

$2\mathbf{X} + \mathbf{Y} \xrightarrow{k} \mathbf{P}$

the rate of reaction is $\frac{d[\mathbf{P}]}{dt} = k[X]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution. At 50 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is(are). **(Use : ln 2 = 0.693)**

(A) The rate constant, k, of the reaction is 13.86×10^{-4} s^{-1}.

(B) Half-life of X is 50s.

(C) At 50 s, $-\frac{d[\mathbf{X}]}{dt} = 13.86 \times 10^{-3}$ mol L^{-1} s^{-1}.

(D) At 100 s, $-\frac{d[\mathbf{Y}]}{dt} = 3.46 \times 10^{-3}$ mol L^{-1} s^{-1}.

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ANSWER KEY

EXERCISE # S-I

1.	(i) 9.00 (ii) 36.00 (iii) 54.00	2.	8.00
3.	x = 83.33 ; y = 120.00	4.	27.00
5.	(a) 0.02 (b) 0.04	6.	8.00
7.	2.00	8.	6.00
9.	0.16-0.17	10.	(i) 7.20 , (ii) 10.00
11.	0.01	12.	60.00
13.	1.20	14.	5.00
15.	(i) 36.00 (ii) 108.00	16.	(i) 0.02, (ii) 70.00
17.	10.00	18.	40.00
19.	0.02	20.	87.5 %
21.	19.00	22.	20.00
23.	0.20	24.	3.00
25.	7500.00	26.	(a) 3.00, (b) 0.75
27.	(i) 1.00 (ii) k = 0.01 (iii) 73.00	28.	(i) 0.00 (ii) 5.00
29.	0.00	30.	1.00
31.	4.00	32.	2.00
33.	0.01	34.	0.06 - 0.07
35.	240.00	36.	0.01
37.	50.00 , 0.18	38.	0.10
39.	300.00	40.	3.00
41	1.54 - 1.55	42.	86.62 - 86.63
43.	36.00	44.	4.00
45.	6.00	46.	5.00
47.	13.44	48.	10.28
49.	19.68	50.	20.00
51.	10.00	52.	150.00
53.	840.00	54.	38.24
55.	20.00	56.	3.00
57.	2.00	58.	1.00
59.	2.00	60.	1.50

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EXERCISE # S-II

1.	**383.00 mm Hg , 404 min.**	**2.**	**50**
3.	**20 min**	**5.**	**(a) 9.24 kcal/mole, (b) 25.60 hour**
6.	**$k = 0.03\ min^{-1}$**	**8.**	**3**
9.	**1440 sec**	**10.**	**5**
11.	**200 min**	**12.**	**3**
13.	**0**	**14.**	**0.04**
15.	**8**	**16.**	**14**
17.	**1680**	**18.**	**0**
19.	**8**		

EXERCISE # O-I

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Ans.	B	D	C	B	C	C	A	C	D	C
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Ans.	A	D	A	C	A	C	D	D	A	D
Que.	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
Ans.	B	A	D	A	D	C	A	A	C	C
Que.	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
Ans.	C	C	B	C	C	A	D	C	A	C
Que.	**41**	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
Ans.	A	B	D	D	C	C	D	B	C	D
Que.	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
Ans.	D	A	B	B	C	A	B	B	C	D

EXERCISE # O-II

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Ans.	A,B,D	A,D	A,C,D	A,B,C	C,D	A,C	C	A,D	A,B	A,B,D
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Ans.	A	C	A,B,C	C	C	C	A	B	A	B
Que.	**21**	**22**	**23**	**24**						
Ans.	C	C	C	A-S	B-R	C-P	D-Q			
Que.	**25**				**26**	**27**	**28**			
Ans.	A-P,Q	B-S,P	C-R	D-P	A	B	C			

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EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	3	4	2	4	2	3	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	1	4	4	1	3	3	4	3
Que.	21	22	23	24	25	26				
Ans.	108	81	26	14	106	45				

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	9	D	B	8	B	B,C,D	B,D	A,D	6.75
Que.	11	12	13							
Ans.	2.30	A	B,C,D							

ALLEN®

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